

Progress Report

Day 8 — API Verification & Backend Stabilization

Task Management MERN Stack Platform

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1. Day-8 Objective

The goal for Day 8 was to finalize and verify the authentication system, Team CRUD APIs, role-based authorization flow, and organize backend API testing using Postman collections.

2. Tasks Completed

Authentication Verification

- Retested JWT authentication endpoints:
 - register
 - login
 - /me

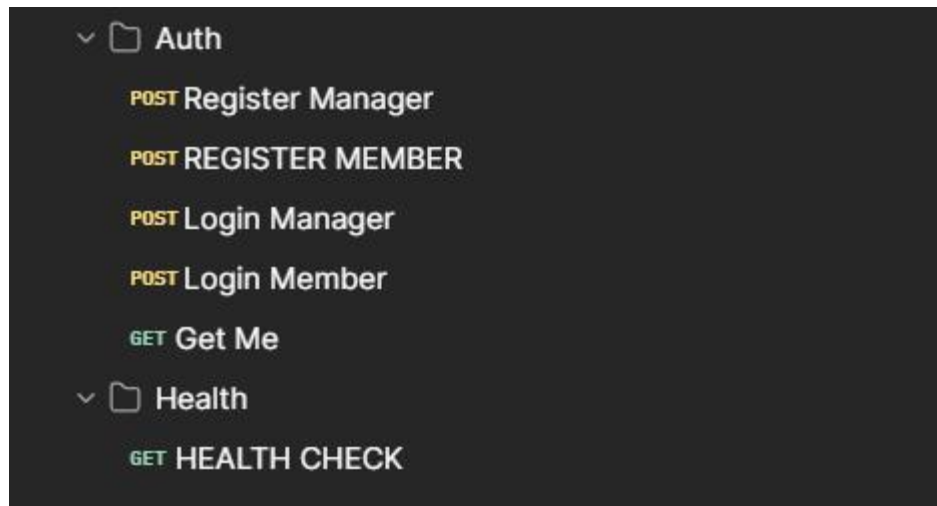


Figure 2.1 — Authentication endpoints verified successfully using Postman.

Team CRUD Verification

- Retested all Team CRUD endpoints:

- create
 - get
 - update
 - delete
- Verified manager-only authorization for protected operations

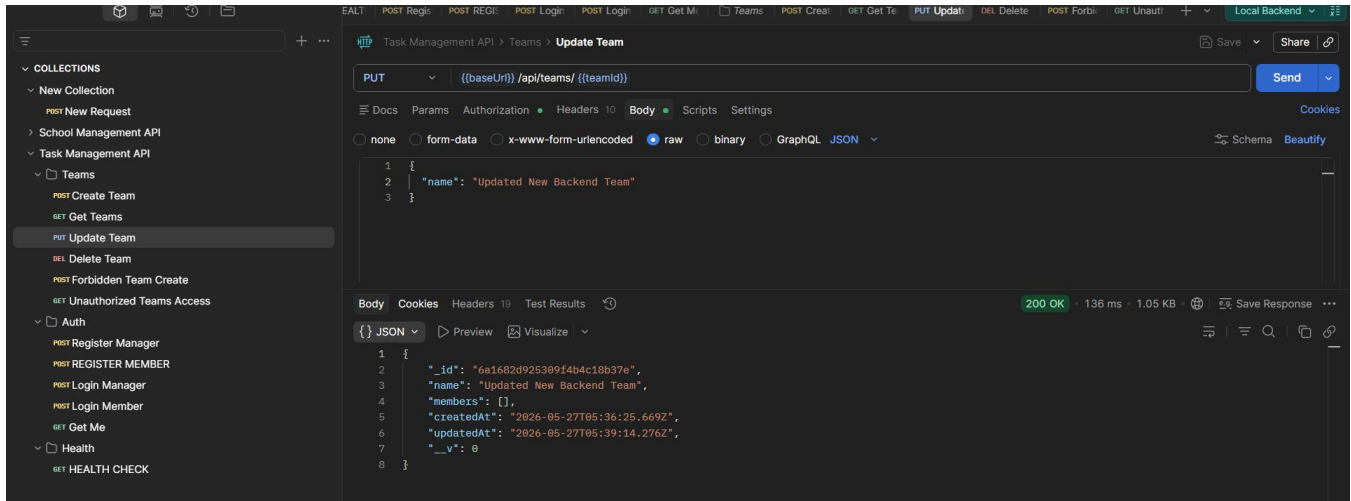


Figure 2.2 — Team CRUD API requests organized inside Postman collection.

Role-Based Access Verification

- Verified manager users can access protected Team CRUD operations
- Verified member users are blocked from manager-only routes using 403 Forbidden responses
- Verified unauthenticated requests return 401 Unauthorized

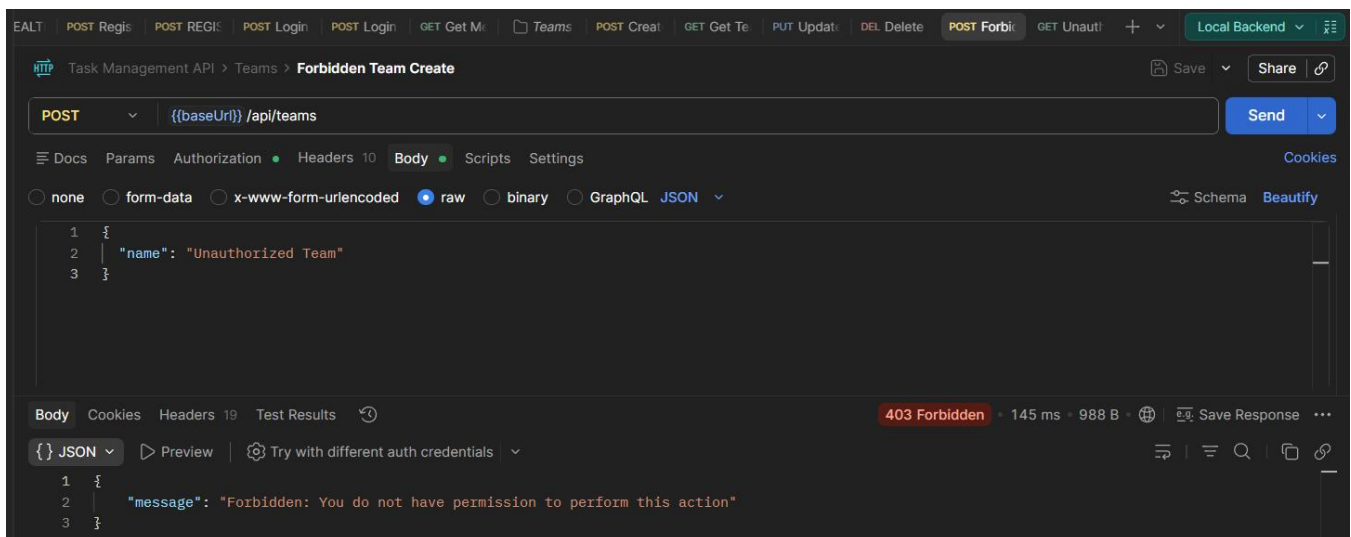


Figure 2.3 — Manager-only route restriction verified successfully.

Postman Collection Organization

- Created centralized Postman API collection for backend testing
- Configured reusable environment variables:
 - baseUrl
 - managerToken
 - memberToken
 - teamId
- Organized requests into:
 - Auth
 - Teams
 - System

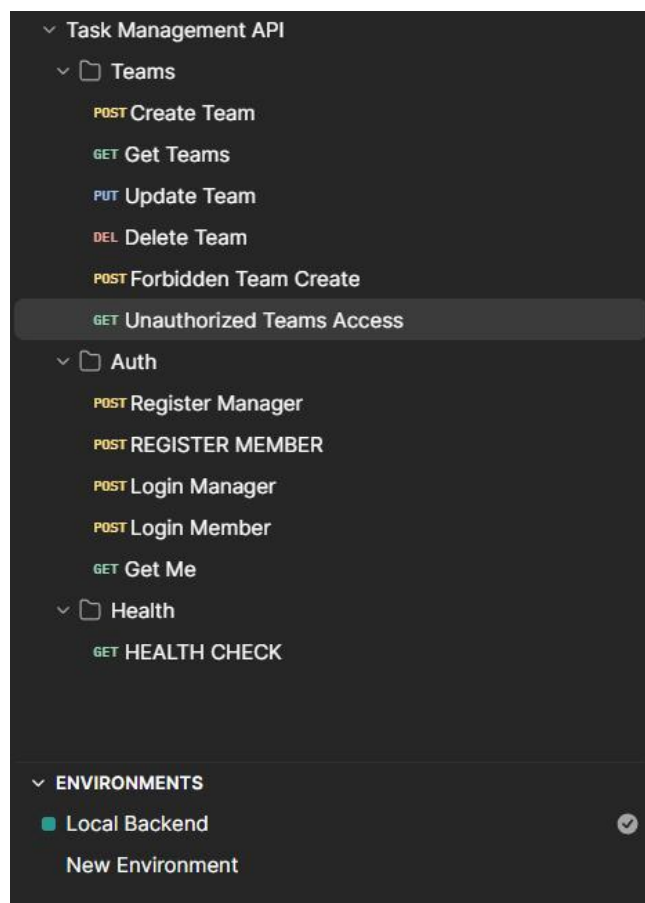
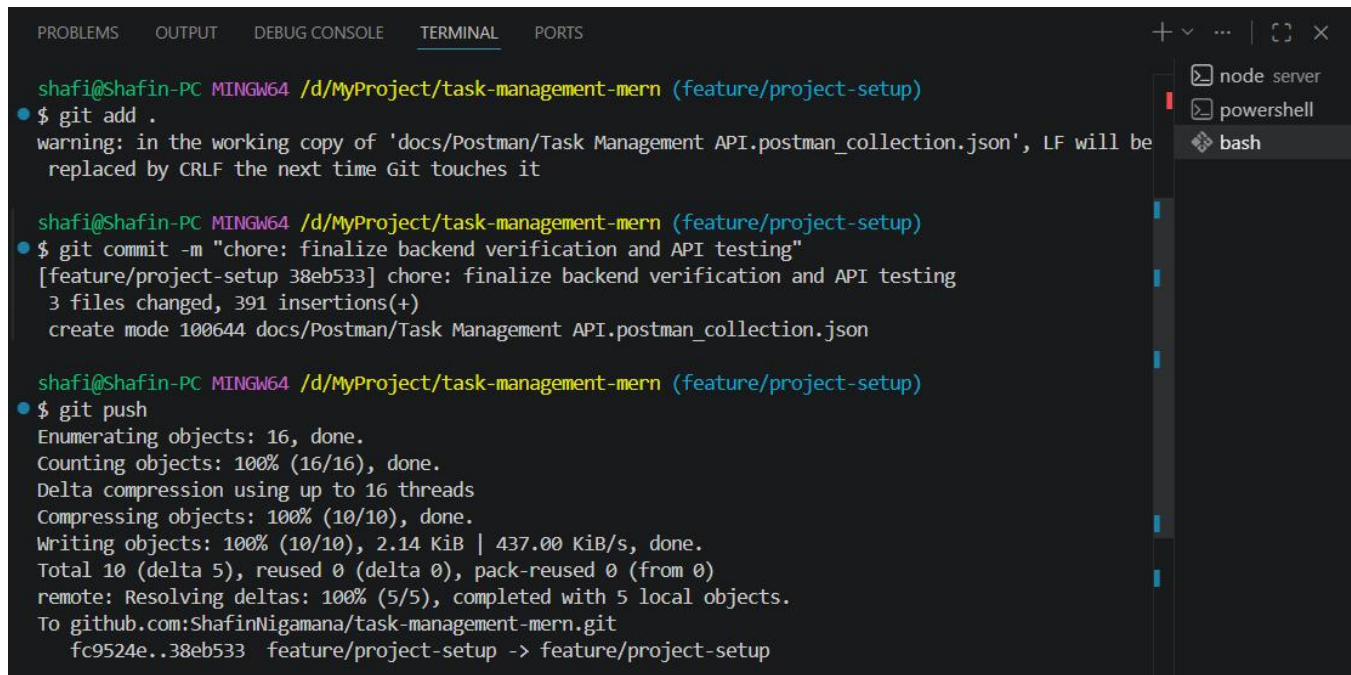


Figure 2.5 — Organized Postman collection for backend API testing.

Repository Update

- Committed backend stabilization and API verification updates using Git workflow
- Pushed latest verification changes to GitHub repository

A screenshot of a Visual Studio Code terminal window. The terminal shows a series of Git commands and their outputs. The prompt is 'shafi@Shafin-PC MINGW64 /d/MyProject/task-management-mern (feature/project-setup)'. The commands executed are: '\$ git add .', '\$ git commit -m "chore: finalize backend verification and API testing"', and '\$ git push'. The output for the push command shows the progress of enumerating, counting, compressing, and writing objects, and finally resolving deltas to push the changes to the remote repository. The output for the push command is: 'Enumerating objects: 16, done. Counting objects: 100% (16/16), done. Delta compression using up to 16 threads Compressing objects: 100% (10/10), done. Writing objects: 100% (10/10), 2.14 KiB | 437.00 KiB/s, done. Total 10 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0) remote: Resolving deltas: 100% (5/5), completed with 5 local objects. To github.com:ShafinNigamana/task-management-mern.git fc9524e..38eb533 feature/project-setup -> feature/project-setup'. The terminal window has tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL', and 'PORTS'. On the right side, there is a sidebar with icons for 'node server', 'powershell', and 'bash'.

```
shafi@Shafin-PC MINGW64 /d/MyProject/task-management-mern (feature/project-setup)
• $ git add .
warning: in the working copy of 'docs/Postman/Task Management API.postman_collection.json', LF will be
replaced by CRLF the next time Git touches it

shafi@Shafin-PC MINGW64 /d/MyProject/task-management-mern (feature/project-setup)
• $ git commit -m "chore: finalize backend verification and API testing"
[feature/project-setup 38eb533] chore: finalize backend verification and API testing
3 files changed, 391 insertions(+)
create mode 100644 docs/Postman/Task Management API.postman_collection.json

shafi@Shafin-PC MINGW64 /d/MyProject/task-management-mern (feature/project-setup)
• $ git push
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 16 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (10/10), 2.14 KiB | 437.00 KiB/s, done.
Total 10 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), completed with 5 local objects.
To github.com:ShafinNigamana/task-management-mern.git
fc9524e..38eb533 feature/project-setup -> feature/project-setup
```

Figure 2.6 — Day 8 backend verification changes pushed to GitHub.

3. Next Steps (Day 9)

- Begin Task schema design using Mongoose
- Implement Task CRUD endpoints
- Add filtering support for tasks
- Prepare audit log integration planning

4. Conclusion

Day 8 focused on final verification and stabilization of the backend authentication and Team management system. Authentication flows, Team CRUD operations, and role-based authorization were verified successfully using organized Postman collections.